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Learning to evaluate: An intervention in civic online reasoning

Sarah McGrew

University of Maryland, College Park, USA

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ABSTRACT

Students turn to the Internet for information but often struggle to evaluate the trustworthiness of what they find. Teachers should help students develop effective evaluation strategies in order to ensure that students have access to reliable information on which to base decisions. This study reports on the results of an attempt to teach students to reason about online information. Students were taught strategies for evaluating digital content that were based on the practices of professional fact checkers.

Eight lessons were devoted to teaching students strategies to effectively evaluate digital content. Pre- and posttests, each composed of four brief, constructed-response items, were administered to 68 11th-grade students who participated in the study. Students' scores improved significantly from pre- to posttest on three of the four tasks: students demonstrated an improved ability to investigate the source of a website, critique evidence, and locate reliable sources during an open Internet search. These results are promising and suggest that explicit instruction on fact-checking strategies may help students develop more effective online evaluation strategies.

1. Introduction

Young people often turn to the Internet for information but do not always effectively evaluate the content they find. Students from middle school through college, in dozens of U.S. states, and in a diverse array of schools have demonstrated misconceptions about how to judge digital information (McGrew, Breakstone, Ortega, Smith, & Wineburg, 2018; Wineburg, McGrew, Breakstone, & Ortega, 2016). Why do students struggle?

One explanation is that the web's growing complexity has outpaced educational efforts. Such efforts may be limited to special visits from a school librarian or free-standing media literacy units. Teachers may give students lists of approved web resources or, if they do allow access to the open web, provide 20- or 30-item checklists to help students evaluate what they find (e.g., Common Sense Media, 2012; Media Education Lab, n.d., News Literacy Project, n.d.). Research-based approaches to integrating instruction in digital evaluations in ways that create measurable changes in student learning are needed. Otherwise, adults cannot expect students to perform complex online evaluations that students have not had support or opportunities to learn.

This study taught students an approach to evaluating digital content based on the practices of professional fact checkers and used measures of students' digital evaluations to assess student learning. It asked, How do lessons on online evaluation skills affect students' abilities to evaluate online information?

E-mail address: mcgrew@umd.edu.

1.1. Evaluating online information

Research has extensively documented how students evaluate online information. When conducting open searches, students relied on the order of search results as a signal of a website's reliability. Students often clicked on the first or second result and expressed the belief that, the higher a site was listed in the results, the more trustworthy it was (Gwizdka & Bilal, 2017; Hargittai, Fullerton, Menchen-Trevino, & Thomas, 2010; Pan et al., 2007). While searching for information about online news sources, college students expressed misconceptions about how Google's Knowledge Panels are curated and often concluded that sources were reliable if they had strong social media presences in the search engine results page (Lurie & Mustafaraj, 2018). Once on webpages, students continued to engage in problematic evaluation behaviors. They rarely judged information based on its source (List, Grossnickle, & Alexander, 2016; Walraven, Brand-Gruwel, & Boshuizen, 2009). Instead, students focused on how closely information matched what they were searching for, which should inform a decision about whether to use information but not whether that information is reliable, and on the appearance of the website (Barzilai & Zohar, 2012; Coiro, Coscarelli, Maykel, & Forzani, 2015; List et al., 2016).

Students ignored sources and evaluated websites based on surface features when conducting searches on relatively straightforward questions (e.g., "Does Microsoft Word store information about the author of a document?" [Hargittai et al., 2010]; "Should we consume food that is out of date?" [Brand-Gruwel, Wopereis, & Vermetten, 2005]). Students fared worse when the content was more contentious. Analyses of thousands of responses to tasks assessing students' ability to evaluate social and political information online showed that students did not distinguish between traditional news and sponsored content and rarely based evaluations on the reliability of a source. Instead, they were swayed by information that appeared to present strong evidence and evaluated websites based on their design or how authoritative they appeared due to features like the presence of a logo or references (McGrew et al., 2018; Wineburg et al., 2016).

Many solutions have been proposed to help people find reliable information and identify misinformation online. Proposals include adding additional source and contextual information to search engine results pages (Schwartz & Morris, 2011) and stories on social media (Hughes, Smith, & Leavitt, 2018), annotating digital content on a range of credibility indicators (Zhang et al., 2018), and directly correcting misinformation when it surfaces on social media (Vraga & Bode, 2017). However, efforts may have unintended consequences (e.g., Pennycook & Rand, 2017) and even risk further entrenching belief in misinformation (Lewandowsky, Ecker, & Cook, 2017). In addition to external tools to identify misinformation, educational efforts should be part of an approach to countering the spread of online misinformation and a way to proactively support people to locate trustworthy content. Indeed, adults reported wanting more support to learn to evaluate online information (Horrigan & Gramlich, 2017) and young people admitted that they struggle to identify fake online content (Robb, 2017).

1.2. Teaching online evaluations

Calls to teach students to evaluate online information have grown louder as worries about students' struggles to evaluate online information intensified (e.g., Bulger & Davison, 2018; Herold, 2016; Shellenbarger, 2016). In a nationally representative survey of 15–27 year olds, Kahne and Bowyer (2017) found that students who reported having media literacy learning opportunities more astutely rated the accuracy of posts containing political arguments and evidence. Additionally, intervention studies suggest that explicit instruction can improve students' abilities to evaluate online information in elementary school (e.g., Zhang & Duke, 2011), middle and high school (e.g., Leu et al., 2005; Walraven, Brand-Gruwel, & Boshuizen, 2013), and college (e.g., Wiley et al., 2009), as well as among adults not in school (e.g., Kammerer, Amann, & Gerjets, 2015). However, these attempts to teach and measure gains in online evaluations have left open questions.

Some interventions fell short in their attempts to teach effective approaches to evaluating the open web. Walraven et al. (2013) investigated an intervention that rewarded students for the number of website features on which they based evaluations. Students' scores increased even when they noted elements that had little or no bearing on a website's reliability (e.g., the age of the website, the amount of information it contained, and whether it was a .com or .org site). Other studies focused on teaching students to analyze the reliability of sources. However, teachers in these studies provided students with information about the source of a website and asked students to draw inferences about its authority, motivations, and overall trustworthiness (Perez et al., 2018; Zhang & Duke, 2011). In another study, Kammerer, Meier, and Stahl (2016) assessed students' memory for and credibility judgments about a set of nine digital sources; students in the experimental group had previously completed a worksheet in which they were prompted to consider the genre of each website (e.g., commercial, scientific, or journalistic). Although these studies suggested that students can learn to reason about relevant features of sources, more research is needed that probes whether students would seek out such information independently—or how they would fare if they attempted to do so on the open web.

A set of studies helped students learn to evaluate digital content but did not directly measure students' abilities to independently judge that content. Research on Internet reciprocal teaching (IRT), which involves students and teachers working together to model, practice, and discuss evaluation strategies like why a site was selected from search results (Leu, Coiro, Castek, Harman, Henry, & Reinking, 2008) showed promising results in middle school classrooms (Colwell, Hunt-Barron, & Reinking, 2013; Leu et al., 2005). However, the measure of students' evaluation skills used in these studies, the assessment of Online Reading Comprehension Ability, explicitly prompted students to investigate the authorship and authority of a website. It did not provide information about whether students could, without specific prompting, engage in accurate evaluations. As Brante and Strømsø (2018) argued in a review of interventions to teach students to engage in evaluating sources in off- and online environments, the measures used in these studies have not illuminated "how students go about searching for source features online or which features they attend to" (p. 792).

Studies of an intervention to teach students to evaluate digital science content also had positive results. Wiley et al. (2009) designed

a computer-delivered intervention based on the acronym SEEK. Materials instructed students to evaluate “(a) the Source of the information, (b) the nature of the Evidence that was presented, (c) the fit of the evidence into an Explanation of the phenomena, and (d) the fit of the new information with prior Knowledge” (p. 1087). Researchers tested the efficacy of similar written materials with high school students (Mason, Junyent, & Tornatora, 2014) and adults (Kammerer et al., 2015) and reported positive changes in students’ ability to rank and select reliable websites. These studies suggested that teaching students to prioritize evaluating sources and evidence helps students develop stronger evaluation tactics. However, what students learn when they are taught to research contentious topics is still an open question.

Despite these existing attempts, the tool at the heart of many widely available approaches to teaching online evaluations is not based on research. This tool, a checklist, focuses on the presence or absence of surface features on a website, such as whether a contact person is provided for the article, if sources of information are identified, and if the spelling and grammar are error-free (e.g., Media Education Lab, n.d.). Questions like these treat evaluation as a relatively straightforward process of checking for specific, often easy-to-manipulate features and summing the number of reliability cues a website gives (Meola, 2004). By focusing students on surface features internal to a website, checklists are likely to lead students in the wrong direction.

1.3. The present study: teaching civic online reasoning

This study tested an approach to teaching students to evaluate online information that was distinct from prior research in several ways.

First, this study focused on teaching students to evaluate social and political information online. As the Internet changes the ways people participate in politics, communicate with fellow citizens, advocate for change, and research issues of social and political importance, schools should consider how to prepare students for these activities. As part of this work, schools should prepare students to “analyze and evaluate information in order to learn about and investigate pressing civic and political issues” (Kahne, Hodgin, & Eidman-Aadahl, 2016, p. 9). This ability to effectively search for and evaluate social and political information online is conceptualized as *civic online reasoning* (McGrew et al., 2018). Thus, this study engaged students in learning to evaluate online content on contentious issues about which there are often no “right” answers or expert consensus.

Next, the evaluation strategies that were the focus of the lessons were based on research with professional fact checkers. Expert studies in domains from physics to history have guided plans for teaching those subjects to novices (Ericsson, Charness, Feltovich, & Hoffman, 2006); the strategies taught in the lessons tested here were based on findings from a study of how professional fact checkers evaluated digital content (Wineburg & McGrew, 2019). Fact checkers successfully evaluated information on contentious topics by prioritizing three questions: (1) Who is behind this information?; (2) What is the evidence?; and (3) What do other sources say? Fact checkers routinely prioritized investigating the source behind a website and used what they learned about its perspective, authority, and potential motivations for presenting the information to make an initial judgment of the source. Next, fact checkers critically evaluated the evidence presented by digital sources, investigating the source of the evidence and how strongly it supported the claim(s) being made. Finally, fact checkers relied on multiple sources of information as they investigated sources and claims.

Fact checkers used a set of strategies to efficiently investigate these questions. To investigate the source behind a website, fact checkers did not stay on a webpage and evaluate it based on surface features like site design, as other studies have shown a wide range of students to do. Instead, fact checkers *read laterally*. Landing on an unfamiliar site, they prioritized finding out more about the site’s sponsoring organization or author. To conduct that research, fact checkers opened new tabs and searched for information about the author or organization outside the site itself. They used news articles, Wikipedia and its references, and fact-checking websites to learn more about the source and its qualifications and potential motivations for presenting the information. Only after fact checkers learned more about the source did they return to the original article. Fact checkers also carefully *analyzed evidence*. They investigated whether evidence was from a reliable source and weighed whether it was relevant to, or directly supported, the claim(s) being made. Analyzing the reliability and relevance of evidence helped fact checkers recognize that hyperlinks or lists of references—even if they included reliable sources—were not necessarily proof that the source itself was trustworthy. Evidence a website put forward to support its claims, including complex data displays, was closely analyzed to check whether data were from reliable sources and actually supported the source’s claims. Finally, fact checkers engaged in *click restraint*. Instead of trusting a search engine to sort pages by reliability, checkers mined URLs and abstracts for clues about each search result, focusing on whether the source was likely to be reliable and what kind of information it might provide. They regularly scrolled down to the bottom of the results page in their quest to make an informed decision about where to click first (Wineburg & McGrew, 2019).

Thus, the lessons were intended to help students learn to evaluate online information on contentious issues by introducing students to civic online reasoning’s core questions and teaching the strategies that fact checkers used to evaluate sources and evidence. In its instructional approach, this study drew on prior research on how students learn complex practices (e.g., Collins, Brown, & Newman, 1989; Greenleaf, Schoenbach, Cziko, & Mueller, 2001; Palinscar & Brown, 1984). Novices learn from expert models and when they are supported to participate in opportunities to apply, co-construct, and revise knowledge through joint activities (Lave & Wenger, 1991). This study hypothesized that student learning would occur through regular opportunities to observe and analyze models of skilled approaches to evaluating online information, practice those approaches in collaboration with peers, and receive feedback and coaching from a teacher. Such approaches, including cognitive apprenticeship and reciprocal teaching, have been found to be effective in helping young people develop complex reasoning and literacy practices (Greenleaf et al., 2001; Palinscar & Brown, 1984).

This study used measures of students’ ability to evaluate social and political information online (McGrew et al., 2018) in order to investigate the questions: How do lessons on online evaluation strategies affect students’ abilities to evaluate online information? How, if at all, do students’ abilities to investigate online sources, evaluate digital evidence, and locate reliable sources change after the online

reasoning lessons?

2. Method

2.1. Participants

This investigation was based in three 11th-grade history classes that were all taught by the same teacher. In the 2016-17 school year, the student body of the comprehensive public high school of which the classes were part was 28% White, 26% Latinx, 24% Asian, 6% Filipino, 3% Pacific Islander, 2% African American, and 11% students who identified as two or more races. Sixteen percent of students were eligible for free or reduced lunch. All students in the three classes were recruited to participate in the study and the lessons and data collection took place during regular class time. Sixty-eight of the 72 students obtained parental consent, assented to participate, and completed both the pre- and posttests. The four students who did not consent to participate in data collection were part of online reasoning instruction during regular class lessons, but pre- and posttest responses were not collected from these students. All participants were either 16 or 17 years old; 35 were female and 33 were male.

The teacher with whom I collaborated was in his 12th year teaching U.S. history. He taught three sections of Advanced Placement U.S. History and was a mentor and coach for other history teachers. Before the study, he reported using computers in most lessons but had not explicitly taught students to evaluate online information. Instead, students used laptops to annotate documents during inquiries, take notes during lectures, record discussions, and videotape presentations. I introduced the teacher to drafts of civic online reasoning lessons in a series of three meetings beginning three months before the intervention. Additionally, we met before and after each lesson to review the sequence of instruction and discuss any questions or modifications to future lessons.

2.2. Instruction in online reasoning

Eight lessons in online reasoning were taught over the course of two months, from early September to early November. These lessons were taught approximately once per week and dispersed among traditional lessons in U.S. history. The lessons were organized into three modules, which corresponded to the three core questions of civic online reasoning: investigating who is behind information, analyzing evidence, and verifying information through multiple sources (see Table 1). Six of the lessons were drafted before the study; two lessons (5 and 8) were developed during the study to provide students with additional practice evaluating online sources and evidence.

In the final lesson, students practiced integrating the core questions and strategies they learned in the preceding seven lessons. Students were given an open Internet research task to conduct in groups. Although scaffolding was provided in the form of a graphic organizer, groups had to select and evaluate websites, analyze evidence, verify sources and arguments, and ultimately defend their answer to the research question.

The lessons included modeling, collaborative group work for guided practice, and class discussions. In the first lesson of each module (Lessons 1, 3, and 6), the teacher modeled a strategy for evaluating online information. He introduced and explained the rationale for using the strategy and demonstrated the strategy while thinking aloud. This allowed students to see the strategy—and the thinking that underlies it—in action. All of the lessons included opportunities for students to engage in collaborative guided practice in small groups. Guiding questions (delivered in paper-and-pencil format or digitally via Google Documents) were provided to support groups as they practiced the strategy. For example, guiding questions while students practiced lateral reading prompted them to identify a website's sponsoring organization and to research that organization outside the site itself. After collaborative group work, the teacher facilitated class discussions. During these discussions, different students shared what they concluded about the question or information they investigated, reviewed what they learned as they practiced the strategy, and tackled additional questions.

The materials used to introduce civic online reasoning strategies and provide opportunities for practice consisted of a range of websites and online posts on social, political, and historical topics. The online content was sometimes relevant to historical topics the class studied; for example, students practiced reading laterally in Lesson 2 with two websites that addressed the question of whether

Table 1
Summary of online reasoning lessons taught.

Module 1: Who is behind this information?	Module 2: What is the evidence?	Module 3: What do other sources say?
<u>Lesson 1:</u> Introduce “Who is behind this information?”; teacher models lateral reading and students engage in collaborative guided practice	<u>Lesson 3:</u> Introduce “What is the evidence?”; teacher models evaluating evidence; students engage in collaborate guided practice evaluating evidence	<u>Lesson 6:</u> Introduce “What do other sources say?”; teacher models analyzing search results and students engage in collaborative guided practice
<u>Lesson 2:</u> Extended collaborative guided practice with lateral reading; class discussion	<u>Lesson 4:</u> Students complete evidence analysis task and examine sample student responses to the task; engage in brief collaborative guided practice	<u>Lesson 7:</u> Collaborative guided practice analyzing evidence and argument of media outlets’ coverage of breaking news event and accompanying class discussion
	<u>Lesson 5:</u> Extended collaborative guided practice evaluating strength of arguments (based on source & evidence); class discussions	
	<u>Lesson 8:</u> Collaborative guided research project integrating learning from all three modules; class discussion	

the Constitution was a revolutionary document. In other lessons, the online content addressed social or political topics that were not directly related to the historical content students studied. For example, in Lesson 3, the class analyzed the evidence presented in a meme about CEO's pay from the organization Occupy Democrats and an infographic on student loan debt without a source.

I observed instruction in two of the three class periods for each online reasoning lesson in order to assess the implementation of the lessons, ensure the lessons were taught consistently across class periods, and inform meetings with the teacher in which we refined the remaining civic online reasoning lessons.

2.3. Measures

The pre- and posttest were each composed of four brief, constructed-response items that were selected in order to assess a range of the domain of civic online reasoning (see Table 2). Each original item had been extensively piloted and think-alouds had been conducted and analyzed in order to check for cognitive validity (McGrew et al., 2018). To avoid testing effects that could result from presenting students with identical tasks on the pre- and posttests, parallel forms of each pretest item were developed ad hoc for the posttest. Stem-equivalent items were designed to target the same core question and strategies for evaluating information as the original item on the pretest (see Fig. 1 for an example of an item on the pretest and its parallel on the posttest). Before the present study, each posttest item was piloted with high school students not part of the study. Responses were analyzed to ensure that they did not substantially differ in form or strategies described from student responses to the original items described in McGrew et al. (2018).

The Ad Identification task provided students with links to two articles—one a traditionally bylined news story and one sponsored content—on the same topic from the same news organization. Students were told to review the articles and asked which was a more reliable source. Successful responses recognized that one article was a paid advertisement by a company and argued that the traditional news story was more reliable.

The Lateral Reading task presented students with an online article and advised that they could use any online resources to determine if it was a reliable source of information. The articles on the pre- and posttest were presented on websites that claimed to be non-profit, non-partisan organizations that sponsored or conducted research. In fact, each was a cloaked website (Daniels, 2009) with connections to corporations with potential conflicts of interest with the topic the article covered—for example, an organization with ties to the beverage industry publishing an article about the negative effects of a soda tax. Students whose responses earned a top score of “2” reported leaving the website and reading laterally to uncover information about the site's sponsoring organization.

The Evidence Analysis item on the pre- and posttest asked students to evaluate questionable photographic evidence (see Fig. 1). Each post makes a claim about what the photograph represents but, in each case, the photograph itself is insufficient evidence for the claim. Students knew nothing about the authority of the person posting the claim (especially given that the photos were uploaded to sites where any user can upload photographs) and had no proof of where the photographs were taken or that the photographs represented what the post claimed they did. Responses that earned a top score of “2” critiqued the evidence based on questions about who took and posted the photo, where it was taken, or whether it represents what it claims it does.

The Claim Research task asked students to research a historical claim with contemporary political ramifications. On the pretest, students investigated the claim that César Chávez, the co-founder of the United Farm Workers union, opposed Mexican immigration to the U.S. On the posttest, the claim under question was that Margaret Sanger, the founder of Planned Parenthood, supported euthanasia.

Table 2

Description of tasks in pre- and posttests.

Assessment Item	Assessment Description	Assessment Question(s)	Core Question Tapped	Strategies & Knowledge Tapped
Ad	Identification	Examine top of two articles (one traditional news story, one sponsored content) and explain which is a more reliable source.	Is Article A or Article B a more reliable source for learning about [topic]?	Who is behind this information?
Identify an advertisement based on its label				
Lateral Reading	Explain whether a website is a reliable source of information.	Q1: Is this a reliable source of information about [topic]? Q2: Explain your answer, citing evidence from the webpages you used.	Who is behind this information? What do other sources say?	Engage in <i>lateral reading</i> about a website's sponsoring organization
Evidence Analysis	Determine whether a photograph posted on social media provides strong evidence for a claim.	Does this post provide strong evidence about [topic]? Explain your reasoning.	What is the evidence?	<i>Analyze evidence</i> by assembling appropriate questions to critique photographic evidence
Claim Research	Research and explain whether a claim is accurate using a search engine and Internet sources.	Q1: Do you believe [claim in question]? Explain using evidence from the websites you consulted. Q2: Explain why the sources you used are strong.	Who is behind this information? What is the evidence? What do other sources say?	Engage in <i>click restraint</i> ; evaluate and explain the reliability of websites (<i>reading laterally</i> when necessary); identify relevant evidence

Students had to judge the accuracy of the claim and explain why the sources they used to reach that judgment were strong. In the case of each claim, an online search does not return straightforward answers. Students needed to sift through websites in order to find sources and evidence they trusted. Successful responses used relevant evidence and provided a sound rationale for why the source it came from was reliable.

Both pre- and posttests were administered digitally via Google Forms. Students worked individually on school computers that had broadband Internet access. They completed each task by typing answers to the prompts and had access to the open Internet while completing all the tasks. The pretest was administered in the history classes in which the lessons were taught two weeks before the first lesson in online reasoning. The posttest was administered two weeks after the final online reasoning lesson, also in history classes. Students had approximately 30 min to complete it during class.

2.4. Analysis

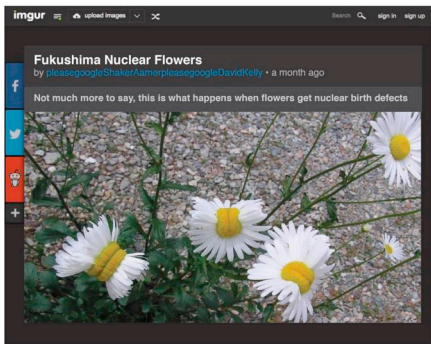
Each assessment item was scored based on rubrics that were drafted over the course of assessment development (see McGrew et al., 2018). Rubrics for three of the tasks (Lateral Reading, Ad Identification, and Evidence Analysis) contained three levels. Because the Claim Research task was more complex, it had a four-level rubric (see Appendix A for all four rubrics). Two trained raters scored all responses on the pre- and posttests. Interrater reliability was as follows: 92% on Ad Identification (Cohen's $Kappa = .86$), 89% on Lateral Reading ($Kappa = .76$), 92% on Evidence Analysis ($Kappa = .86$), and 92% on Claim Research ($Kappa = .88$). Statistical tests determined whether significant changes in performance occurred between the pre- and posttests.

Next, student responses were analyzed and coded based on the strategies that students described using. An initial coding scheme for each item was developed based on extant research on how students evaluate information as well as the most common reasoning strategies observed while developing each task. Codes were added and refined in order to fully represent the range of student reasoning on that task, and all student responses were coded using that scheme (see Appendix B for the final coding scheme for each task). A second rater coded 25% of student responses to each task. Interrater reliability was 91% on the Ad Identification task (Cohen's $Kappa = .87$), 95% on Lateral Reading ($Kappa = .93$), 90% on Evidence Analysis ($Kappa = .86$), and 93% on Claim Research ($Kappa = .90$). Finally, the codes were analyzed in order to describe the variation in reasoning strategies that students used in each task and how, if at all, their reasoning strategies changed from pre- to posttest.

3. Results

Because scores were non-normally distributed, two-tailed Wilcoxon signed-rank tests were used to evaluate changes in scores from pre- to posttest.

On March 11, 2011, there was a large nuclear disaster at the Fukushima Daiichi Nuclear Power Plant in Japan. This image was posted on Imgur, a photo sharing website, in July 2015.



Does this post provide strong evidence about the conditions near the Fukushima Daiichi Power Plant? Explain your reasoning.

The civil war in Syria began in 2011 and continues through the present. This image was posted on Twitter, a social media platform, in January 2014.



Does this post provide strong evidence about conditions for children in Syria? Explain your reasoning.

Fig. 1. Parallel forms of the Evidence Analysis task from the pretest (left) and posttest (right).

3.1. Ad Identification

Ad Identification was the only assessment on which there was no significant improvement in student scores. Student scores on this task averaged 0.76 ($Mdn = 0$; $IQR = 2$) on the pretest and .60 ($Mdn = 0$; $IQR = 1$) on the posttest on a 0–2 scale. Change from pre-to posttest was not significant ($Z = -1.44$; $p = .15$). More than half the students (58% on the pretest and 51% on the posttest) based their explanation of which article was more reliable on the *contents* of the articles. These students argued that features such as the presence of evidence, the amount of information provided, or the argument presented made one article more reliable. Other students (38% on the pretest and 35% on the posttest) successfully discounted one of the articles because it was an ad. These students were divided in the extent to which they identified the potential conflict of interest between the sponsoring organization and the contents of the article. A final group of students (4% on the pretest and 14% on the posttest) focused on differences in dates of publications or the name of the author provided on the news story. Thus, there was no substantial change in student responses or reasoning from pre- to posttest. None of the online reasoning lessons directly addressed the issue of how to identify online advertisements or taught students that some online advertisements are labeled by words like “sponsored content” or “paid post.” It is thus perhaps unsurprising that there was no significant change in scores on this task.

3.2. Lateral Reading

Students’ scores on the Lateral Reading task averaged 0.34 ($Mdn = 0$; $IQR = 0.5$) on the pretest and .75 ($Mdn = 0$; $IQR = 2$) on the posttest, a significant change ($Z = -3.59$; $p < .001$). The growth students demonstrated from pre-to posttest can be characterized by two primary changes. First, more than three times the number of students successfully read laterally on the posttest than on the pretest (9% on the pretest and 32% on the posttest).

Second, there was a substantial shift in the number of students who moved from an approach that did not prioritize investigating the question “Who is behind this information?” to an approach that did prioritize investigating this question. On the pretest, 60% of students did not prioritize investigating the source of information. These students reported analyzing the contents of the article or attempting to fact check a single decontextualized fact from it. Such strategies kept students from discovering the site’s connection to a corporation with a potential conflict of interest with the topic covered in the article. As a result, students erroneously concluded that the website was a reliable source. In contrast, on the posttest, just 10% of students took such an approach. Ninety percent of students investigated who was behind the information in order to determine how reliable the article was. These students reported reading laterally to learn more about the website’s sponsoring organization, investigating the author’s credentials, and visiting the website’s About page. They showed evidence of prioritizing the question “Who is behind this information?” when they encountered unfamiliar web content.

3.3. Evidence Analysis

Students’ mean scores on the Evidence Analysis task were 0.49 ($Mdn = 0$; $IQR = 1$) on the pretest and 1.62 ($Mdn = 2$; $IQR = 1$) on the posttest, a significant change ($Z = -6.23$; $p < .001$).

Students moved from largely accepting the problematic evidence on the pretest to critiquing it on the posttest. Before the online reasoning lessons, 14% of students assembled relevant critiques of a photograph of supposedly “nuclear” daisies. After the lessons, 73% of students successfully critiqued a photograph that purported to show a Syrian child sleeping between his parents’ graves. These students raised questions about who took the picture, who posted it on social media, where the photo was taken, whether it represented what it claimed, or whether it had been photoshopped. For example, one student wrote:

“This post does not provide evidence about conditions for children in Syria. This is a Twitter post posted by an unreliable source with no sourcing on how they got their information or when or where the picture was taken. It is not known if the child is really ‘sleep[ing] between his parents’ or if this picture is even taken in Syria.”

Another group of students accepted the photographs as evidence. These students—64% on the pretest and 11% on the posttest—did not critique the evidence presented in the photo. Finally, some students equivocated in their responses. Twenty-two percent of students on the pretest and 16% on the posttest presented at least one critique of the evidence but ultimately accepted the photograph as evidence for the poster’s claims.

3.4. Claim Research

Students’ mean scores on the Claim Research task grew from 0.86 on the pretest ($Mdn = 0$; $IQR = 2$) to 1.55 on the posttest ($Mdn = 2$; $IQR = 1$); change from pre-to posttest was significant ($Z = -3.77$; $p < .001$).

Students improved in their ability to select reliable websites and to provide strong explanations about why those sources were reliable. On the pretest, 75% percent of students provided no or an irrelevant explanation for why the sources they used were reliable. Such explanations focused on website features unrelated to the authority or trustworthiness of the source, including the website’s date of publication, appearance, or the argument the article made. For example, a student who focused on the date of publication on the pretest wrote, “The sources I used were strong because one of my articles has updated their information.” On the posttest, 29% percent of students provided explanations that focused on features irrelevant to reliability. Instead, 71% of students provided a relevant explanation about the reliability of the source by focusing on its authority or trustworthiness. For example, the student who focused on

the date on the pretest wrote on the posttest, “This source is strong because it is from New York University and they have a whole project based on [Sanger’s] writings and speeches. They also source their work and state where it came from.”

Students’ primary area of struggle on the posttest was selecting sources that provided information directly related to the question of whether Margaret Sanger supported euthanasia. Many students offered explanations based on sources that discussed Sanger’s views on eugenics or cited biographies of Sanger that were silent on the question of euthanasia and concluded that she must not have supported the practice.

4. Discussion

Student scores grew from pre-to post on three of the four online tasks. Students showed evidence of prioritizing finding out more about the backers of an unfamiliar website, they moved from largely accepting problematic evidence on social media to rejecting it, and they more successfully located reliable information on a contentious question in the course of open research. These gains came after just eight lessons in online reasoning taught over the course of one semester. Considering the extensive evidence documenting students’ struggles to evaluate digital information (e.g., [Gwizdka & Bilal, 2017](#); [Lurie & Mustafaraj, 2018](#); [McGrew et al., 2018](#)), this sets a hopeful precedent. The results of this study suggest that students can learn to employ more sophisticated strategies to evaluate sources and evidence online.

The positive changes in student performance reported in this study are in line with prior studies of digital literacy interventions (e.g., [Walraven et al., 2013](#); [Wiley et al., 2009](#); [Zhang & Duke, 2011](#)). Further, this study’s distinct approach contributes to our understanding of how to teach students to evaluate contentious information online. First, this study relied on measures that required students to actually evaluate online information and write descriptions of their judgments. Tasks assessed the extent to which students could, faced with online information and with limited support, effectively make judgments of reliability. Prior studies have relied on measures that asked what students *would* do, provided students with source information that they would, on the open web, need to find themselves, or scaffolded students’ evaluations by prompting them to investigate specific website features (e.g., [Leu et al., 2005](#); [Perez et al., 2018](#)). The tasks on the pre- and posttests in this study hewed more closely to the environment in which students find themselves when they are on the open Internet and trying to decide what to trust.

Next, this study aimed to teach students approaches to analyzing information that skilled online evaluators used ([Wineburg & McGrew, 2019](#)). This was the first attempt to teach these strategies to secondary students. Thus, this study takes a first step in investigating whether the framework and strategies of civic online reasoning, if taught explicitly, can help students develop more skilled approaches to evaluation. Because the approaches to evaluating online information used to craft the lessons were based in understandings of skilled practice, they present a novel opportunity to explore how students learned the strategies and used them to evaluate online information.

Finally, the goal of the lessons was not simply to help students use the Internet to find reliable answers to straightforward questions. The lessons tackled a weightier task: they aimed to help students evaluate complex, even controversial social and political content. Students’ ability to evaluate online information improved on measures that addressed thorny content, including minimum wage and beverage taxes in the Lateral Reading task, nuclear radiation and the Syrian civil war in the Evidence Analysis task, and euthanasia and immigration policy in the Claim Research task. As students turn to the Internet for political information, the civic health of our communities is diminished if students struggle to differentiate high-quality from questionable content. Thus, students need support learning to navigate information they will use to make political choices and take informed action ([Kahne et al., 2016](#)). This study showed that lessons that teach students to evaluate these kinds of information are possible and can result in positive change.

Students showed evidence of developing new strategies to evaluate online information. After the online reasoning lessons, 90% of students prioritized the question “Who’s behind this information?” on the Lateral Reading task. Although many of these students did not fully succeed in the task, their progress showed that students can learn to prioritize investigating a source even when they are not prompted to do so or not explicitly provided with source information. Students also grew in their ability to ask “What’s the evidence?” The largest improvement came on the Evidence Analysis task, where students needed to raise specific concerns about photographic evidence but did not need to leave the post or conduct additional research. Students’ performance on the posttest suggests that students were less willing to simply accept evidence and more able to marshal appropriate critiques of what they saw.

However, results of this study suggest that students need more support to successfully carry out strategies for evaluation. Although they nearly all *attempted* to investigate the source of information in the Lateral Reading task, the majority of students were not fully successful at reading laterally to uncover a conflict of interest. Students need more support and practice with the process of lateral reading to better understand both why the strategy is preferable to other means of investigating a source (e.g., consulting its “About” page) and how to execute lateral reading. Future lessons could also support students to consult a range of sources while reading laterally, teaching them to use reliable news and fact checking organizations or to mine Wikipedia’s references to find reliable sources. Lessons could also help students learn to conduct more efficient searches for information about an author or organization, such as putting an organization’s name in quotation marks or adding the word “funding” or “bias” to a search to uncover more sources.

Students also need more practice with open searches. In the Claim Research task, students successfully searched for more reliable sources; however, they often struggled to find evidence that was relevant. More guided practice would help students learn to balance the priorities of relevance and reliability and to develop flexibility and persistence needed to locate sources that meet both criteria. This would also help students develop greater flexibility and a repertoire of strategies for locating reliable and relevant sources during a search.

4.1. Limitations

Because it investigated a new curricular approach, this study was based on data from a relatively limited sample—68 students from a single school site being taught by an experienced teacher. Additionally, a control group that did not participate in the online reasoning lessons was not included in the study. Students' gains in online reasoning from pre-to posttest suggest that the effects would persist with a more complex experimental design and a larger sample that included diverse school contexts, but this claim should be tested in future studies (see section 4.2).

The measures used in this study relied on students' written reports of how they evaluated information. The measures did not track students' online movements or record the websites they opened. Measures that recorded online movements would offer additional evidence about students' evaluation strategies. Further, the measures focused on assessing students' use of strategies for evaluating online information. Additional measures that tapped students' dispositions and knowledge, particularly of Internet sources and of tactics for navigating web searches, would provide further insight into the effects of the lessons. Although the measures included materials on contentious social and political topics, the pre- and posttests did not probe students' prior knowledge or beliefs about these topics. Because judgments about information on contentious issues are mediated, often unconsciously, by our feelings and beliefs (Kunda, 1990; Lodge & Taber, 2013), this is another limitation of the study.

Finally, counterbalancing tasks on the pre- and posttests may improve confidence in the findings. Parallel items were used in order to avoid testing effects; giving half of students one form on the pretest and its parallel form on the posttest (while the other half of students received the opposite) would help ensure that any differences in parallel tasks were not responsible for the changes reported. Still, the results conformed to what one would expect to see if the lessons had an effect. There was significant growth from pretest to posttest on three tasks that tapped strategies covered in the lessons: lateral reading, analyzing evidence, and click restraint. There was no significant change in student performance on Ad Identification, a task that assessed an aspect of investigating sources—how online ads are labeled—that was not directly covered by the lessons. On the tasks that tapped strategies covered in the lessons, not only did students' mean rubric scores increase, but their strategies for evaluation shifted from pre- to posttest.

4.2. Future research

Future studies can build on this research by investigating the lessons themselves and by developing and testing additional measures of student learning. Such studies should probe how teachers with varying amounts of experience deliver lessons in online reasoning and what students learn as a result of the lessons shortly after they are taught and on a delayed posttest. Further, such studies should make use of quasi-experimental design in order to test the efficacy of these lessons in comparison to no instruction in online reasoning, a different number of lessons in online reasoning, or different approaches to teaching students to evaluate online information like checklists. Such studies should include formal implementation fidelity evaluations.

Future studies should also develop and test additional measures of students' online reasoning. These studies could formally test the parallel forms of measures as well as make use of more varied and complex measures. They could also make use of more widely varying topics in order to address issues of student motivation to engage in productive evaluations (List & Alexander, 2017; Metzger, 2007) and ask students to report on their prior knowledge and/or beliefs on the topic and level of engagement with the tasks. Information yielded by such measures would help us better understand students' developing online reasoning when they evaluate contentious digital information.

4.3. Conclusion

In the last few years, a great deal of worry has surfaced about how students—and the rest of us—struggle to evaluate information. This worry is well founded: if citizens make decisions that are not based on trustworthy information, there are real consequences for our communities and countries. And yet, the worry has not been accompanied by concerted efforts to teach students research-based strategies for evaluating online information. This study attempted to begin this work. Its findings, that young people do show evidence of significant growth in their ability to evaluate unfamiliar websites, critique digital evidence, and locate reliable sources during an open search, are simultaneously obvious and hopeful. If young people are taught to evaluate online information through lessons that are thoughtfully crafted, they may show evidence of conducting more sophisticated evaluations. In this way, this study lays the foundation for continued work to improve lessons and test their effectiveness with larger, more diverse samples of students. A citizenry that more thoughtfully navigates and evaluates the online information ecosystem is possible, and teachers can be leaders in these efforts.

Appendix A. Assessment Rubrics

Ad Identification Rubric

Description	Sample Student Response
2 Student identifies that one article is sponsored by a company with a vested interest in the article's topic. Student provides a clear rationale for why this makes the article less reliable	"Article A [is more reliable because it] is not sponsored or affiliated with any other organization and therefore more accurately represents a reasoned opinion with no underlying motive that another company might hold. Article B is sponsored by Shell, a gas company whose motive may be to get rid of the thought of climate change being real for business purposes."
1 Student identifies one article as sponsored content and explains that this makes it less reliable as a source, but the explanation is limited (e.g., the student does not explain the conflict of interest in the ad or why it is problematic)	"Article B is sponsored by Shell, so it is probably biased."
0 Student does not identify the sponsored content as a relevant consideration or identifies the sponsored content but argues that it is the more reliable source	"Article B [the sponsored content] is more reliable. Article B talks more about ways to stop climate change and what can be used to help gain clean energy. In Article A, it talks more about the politics on the use of clean energy and less on policies."

Lateral Reading Rubric

Description	Sample Student Response
2 Student rejects the website as a reliable source and provides a clear rationale based on a thorough evaluation of the organizations behind the website	"I learned (www.sourcewatch.org/index.php/American_Council_on_Science_and_Health) that the ACSH was funded by large scale corporations such as Kelloggs, Pepsico, and the American Beverage Association. It would be in their best interests to not have their products taxed and bring sales down, and since they fund this website, they have a lot of sway in what is published."
1 Student rejects the website as a reliable source and identifies the intent of the website's sponsors but does not provide a complete rationale (e.g., states they are biased but does not identify industry ties)	"No, this is not a reliable source of information. Upon my further investigation, I realized that this 'American Council for on Science and Health' is AGAINST taxing foods/drinks such as soda, in this case (https://en.wikipedia.org/wiki/American_Council_on_Science_and_Health)."
0 Student accepts the source as trustworthy or rejects the source based on irrelevant considerations (e.g., bases evaluation on the website's appearance, contents of article, level of agreement with article, or "About" page description)	"The About page stated that their mission is to 'to publicly support evidence-based science and medicine.' This means that ACSH bases their arguments off of facts."

Evidence Analysis Rubric

Description	Sample Student Response
2 Student argues the post does not provide strong evidence and questions the source of the post (e.g., we don't know anything about the author of the post) and/or the source of the photograph (e.g., we don't know where the photo was taken)	"We have no evidence to even know if this picture was taken near the Fukushima Power Plant, due to this we cannot use this picture to show the effects of the disaster. Also, the poster is a random Imgur user, we have no idea who this person is or even if they took the picture."
1 Student argues that the post does not provide strong evidence, but the explanation does not consider the source of the post or the source of the photograph, or the explanation is incomplete	"This image does not really provide strong evidence about the conditions near the power plant because there could be a lot more factors that could affect the way daisies grow. It does provide an interesting perspective but it is not strong evidence because some of the daisies in the picture look completely fine."
0 Student argues that the post provides strong evidence or uses incorrect (e.g., based on the quality or appearance of the photo) or incoherent reasoning	"This post does provide strong evidence about the conditions near the Fukushima Daiichi Power Plant, as it provides an example of birth defects that result from radiation. Readers are led to believe that if the conditions can affect flowers, it can also affect humans."

Claim Research Rubric

Description	Sample Student Response
3 Student constructs an explanation based on a reliable source ^a that provides information relevant to the claim under question. Student provides a clear rationale for why the source is reliable	Q1: "I don't believe Cesar Chavez opposed Mexican immigration on the whole, but I do believe he opposed illegal immigration. He fought for policy reform, arguing that the use of undocumented workers would undermine the migrant workers by providing cheap labor, thus eliminating the need to pay (continued on next page)

(continued)

Description	Sample Student Response
2 Student constructs an explanation based on a reliable source and provides a clear rationale for why the source is reliable. However, the source does not provide information that is directly relevant to the claim under question	migrant workers higher wages.” Q2: “I used abcnews.go.com/ABC_Univision/Politics/cesar-chavez-complex-history-immigration/story?id=19083496 . This is from ABC News, a well-known, reliable news organization.” Q1: “I believe that Chavez opposed Mexican immigration to the U.S., because he believed they wouldn’t benefit farm worker justice, which he fought for and is remembered by.” Q2: “The National Public Radio is a non-profit membership media organization that focuses on news and cultural programming, and it is generally considered to be a reliable source. (http://www.npr.org/2016/08/02/488428577/cesar-chavez-the-life-behind-a-legacy-of-farm-labor-rights)”
1 Student constructs an explanation based on a reliable source but provides no rationale or an irrelevant rationale for why the source is reliable (e.g., the website’s appearance, length of article, level of agreement with article, or “About” page description)	Q1: “I believe that Cesar Chavez opposed illegal immigrants because they interfered with his union. Illegal immigrants will work for extremely low wages and when Cesar Chavez would organize strikes, it would not work out because the illegal immigrants would come and work for lower wages.” Q2: “ http://abcnews.go.com/ABC_Univision/Politics/cesar-chavez-complex-history-immigration/story?id=19083496 ”
0 Student constructs an explanation based on an unreliable or overtly partisan source and provides no rationale or an irrelevant rationale (e.g., the website’s appearance, length of article, level of agreement with article, or “About” page description) for selecting the source	Q1: “I believe Cesar Chavez opposed Mexican immigration to the US. According to spectator.org , Chavez was constantly using words like ‘wet backs,’ and ‘illegals’ when he was being questioned for being unreasonable and hypocritical. He wanted to protect his Union and didn’t want it to be affected by illegal workers or immigrants.” Q2: “It pulls from multiple different sources and seems fairly accurate. https://spectator.org/59956/cesar-chavez-anti-immigration-his-union-core/ ”

^a Reliable sources were defined as those that had well-established research or journalistic credentials themselves and/or accurately cited sources with established credentials. Such sources had authors with professional backgrounds in journalism or history and processes in place to ensure the accuracy of their materials (e.g., editors, fact checkers, and avenues to issue corrections when necessary).

Appendix B. Assessment Codes

Ad Identification

Code	Subcode	Description
Identify advertisement	Describe conflict of interest	Identify that one article is an advertisement by a company with a potential conflict of interest with the topic of the article
	Do not describe conflict of interest	Identify that one article is an advertisement by a company, but do not explain the potential conflict of interest with the topic of the article
Evaluate other source features		Explain selection of article based on date of publication or the author of the traditional news story
Evaluate Contents	Evidence	Explain selection of article based on the presence of evidence (e.g., hyperlinks, quotes, graphs, or statistics)
	Amount of information	Explain selection of article based on the length of the articles or the amount of information provided
	Argument of article	Explain selection of article based on the argument presented in one or both

Lateral Reading

Code	Subcode	Description
Prioritize investigating source of information	Successful lateral reading	Locate accurate information about the website’s sponsoring organization and their potential conflict of interest with the topic at hand outside the site itself
	Unsuccessful lateral reading	Report leaving the website to investigate the source but do not locate accurate information about the website’s sponsoring organization and their potential conflict of interest with the topic at hand
	Source within site	Read about the author or organization responsible for the article within the website itself (e.g., on a section of author biographies or the “About” page)
Do not prioritize investigating source of information	Contents	Evaluate the contents of the article, including its length, the appearance of evidence or links, or the perceived accuracy of the information
	Discrete fact	Check accuracy of a fact or data point included in the article outside the site itself

Evidence Analysis

Code	Subcode	Description
Critique evidence	Photographer	Raise question that there is no information about who took the photograph
	Social media	Raise concern that anyone can post a photograph on social media and it is not required to be accurately portrayed
	Location	Raise question that there is no proof that the photograph was taken where the poster claims it was
	Causation	Raise question that there is no proof that the cause the poster claims is at play (e.g., nuclear radiation or the death of parents in the civil war) is in fact true
Accept evidence	Photoshopped	Raise concern that any photograph can be photoshopped
	Not enough damage	Explain that more photos are needed (of other plants, animals, etc.) but still express belief in the poster's claim about the photo
	Visual evidence	Explain that the poster's claim can be seen in the photograph
	Background knowledge	Explain that the poster's claim is true based on background knowledge (e.g., how radiation can affect living organisms or the impacts of the Syrian civil war)
Equivocate		Response is coded as including both critiques of the evidence and acceptance of the evidence

Claim Research

Code for Answer to Q2 (Explain why the sources you used are strong)	Subcode for Answer to Q2	Description
No or irrelevant explanation for reliability of source	No explanation	No explanation is provided for why the source is strong or why it was selected.
	Content	Evaluate the contents of the article, including its length, inclusion of evidence, or the perceived accuracy of the information
	Date	Evaluate the date the article was written or posted
	Official appearance	Evaluate the appearance or seeming authority of the website based on features like an official-sounding name, the presence of a listed author name, or web design
Relevant explanation for reliability of source	Strength of source	Evaluate the source of the website (focusing on features like its authority and reputation for trustworthiness) to explain why it is likely to be a reliable source of information

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